

Project Work Plan

Course Name: Software Development I

Course Code: CSE 2216

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Project Title: Treasure Hunt

Course: Software Development Lab

Programming Language: JavaScript

10-Week Work Plan for Treasure Hunt Project

Week	Phase
1	Project Planning & Concept Finalization
2	Requirements Analysis & System Design
3	Environment Setup & Project Initialization
4	Core Rendering & Canvas Setup
5	Procedural Island & Player Movement
6	Items, Inventory & Torch Mechanic
7	Riddles, Scoring & Mini-Map
8	Mobile Support & UX Polish
9	Error Handling & Testing
10	Final Testing, Documentation & Submission

Week 1: Project Planning & Concept Finalization

- Select project topic and objectives.
- Define gameplay mechanics and project scope.

Prepare initial project plan.

Week 2: Requirements Analysis & System Design

- Gather functional requirements for gameplay and UI.
- Design system architecture and file/module layout.
- Create basic wireframes for HUD and menus.

Week 3: Environment Setup & Project Initialization

- Install Node.js and configure the development environment.
- Set up project structure (src/, components/, utils/, styles/).
- Initialize package.json and local dev server with live reload.

Week 4: Core Rendering & Canvas Setup

- Set up the HTML5 Canvas and core game loop (game.js).
- Implement the rendering pipeline (renderer.js).
- Apply base styling and theme (main.css, game.css).

Week 5: Procedural Island & Player Movement

- Implement procedural island map generation (map.js).
- Add player movement and state handling (player.js).
- Wire up WASD / Arrow Key controls.

Week 6: Items, Inventory & Torch Mechanic

- Implement coins, clue scrolls, and treasure chest logic (items.js).
- Add inventory toggle (I) and torch fuel countdown system.
- Implement darkness/game-over condition when torch expires.

Week 7: Riddles, Scoring & Mini-Map

- Add the 5 clue scroll riddles guiding players to the treasure.
- Implement scoring based on coins collected and time taken.
- Add mini-map toggle (M) and pause functionality (ESC).

Week 8: Mobile Support & UX Polish

- Implement on-screen D-pad for mobile/touch devices.
- Add settings and leaderboard persistence via localStorage (storage.js).
- Polish menus, panels, and in-game overlays.

Week 9: Error Handling & Testing

- Test procedural map generation across multiple runs.
- Handle edge cases (torch reaching zero, invalid movement, empty inventory).
- Fix bugs and optimize rendering performance.

Week 10: Final Testing, Documentation & Submission

- Perform complete gameplay testing across devices.
- Prepare README, screenshots, and design notes (docs/).
- Finalize and submit the project.